

Catastrophe as Genesis: A Thermodynamic Framework for Biosphere Coherence Across 3.8 Billion Years, with an Exploratory Counterfactual Suggesting Chicxulub May Have Opened a Narrow Path to Meta-Cognitive Consciousness

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Abstract

We introduce $B(t)$, a continuous biosphere coherence index spanning 3.8 Ga (the origin of life) to the Neolithic Revolution, computed as the geometric mean of six normalized dimensions: biodiversity, complexity, resilience, network integration, energy throughput, and information capacity. Calibrated against Sepkoski’s marine genus diversity curve (Pearson $r = 0.92$, $R^2 = 0.85$, $n = 37$), $B(t)$ is grounded in published paleontological data (PBDB, GEOCARB, $\delta^{13}\text{C}$ isotopic records) and employs five methodological innovations: taphonomic SQS-style Bayesian correction for preservation bias, Lotka-Volterra niche-filling recovery dynamics, Kleiber metabolic scaling for network integration, $\delta^{13}\text{C}$ geochemical bounding of Precambrian confidence intervals, and a shared thermodynamic currency (HANPP) bridging biosphere coherence to civilizational indices.

Using a novel **Complexity Cap** counterfactual method—in which suppressed extinction events lock the biosphere at its pre-extinction thermodynamic ceiling—we find that mass extinctions can be *generative* under assumptions of strong evolutionary lock-in. The strength of this result depends critically on the counterfactual regime assumed. Under a hard

cap (permanent lock-in), all six tested events appear generative; under a soft cap (10% innovation leakage) the result holds for most events; under delayed escape (30 Ma hold then release) or no cap, it does not.

We extend the framework to an exploratory consciousness-feasibility analysis. Under the two more restrictive cap regimes (hard and soft), the Chicxulub impact appears necessary for meta-cognitive consciousness: feasibility peaks at 0.24–0.27, below the 0.30 threshold. Under weaker assumptions, meta-cognition emerges regardless. The verdict is thus **model-dependent**: whether the asteroid was necessary for consciousness depends on whether evolution can innovate past an incumbent apex regime without catastrophic clearing—itself an open question in paleobiology.

Keywords: biosphere coherence, mass extinction, counterfactual, consciousness emergence, thermodynamic framework, deep time, Chicxulub

Introduction

The history of life on Earth is punctuated by catastrophic mass extinctions that destroyed 35–96% of genera within geologically brief intervals [1–3]. Traditional treatments frame these events as interruptions of an otherwise progressive trajectory—losses that the biosphere must recover from before resuming diversification [4, 5]. We propose an alternative framework: catastrophe is not merely a perturbation but a *generative* force that produces higher eventual coherence by clearing evolutionary local maxima and enabling niche radiation into previously inaccessible thermodynamic configurations.

This claim requires a quantitative coherence measure that spans the full history of life. Previous diversity indices—Sepkoski’s compendium [1], Rohde and Muller’s cycles analysis [6]—track genus counts without integrating the multi-dimensional aspects of biosphere organization: complexity, resilience, energy throughput, network structure, and information capacity. We introduce $B(t)$, a six-dimensional biosphere coherence index that captures these properties from 3.8 Ga (the earliest evidence of life [7]) through

the Neolithic transition ($\sim 10,000$ BCE).

The framework connects to civilizational coherence via a shared thermodynamic currency: Human Appropriation of Net Primary Productivity (HANPP) [8]. As human civilization co-opts biosphere energy throughput through land use change and fossil carbon extraction, $B(t)$ is mechanically depressed while civilizational coherence $K(t)$ rises. The unified index $C(t) = B_{\text{HANPP}}(t) + K(t) \times (1 - B_{\text{HANPP}}(t))$ produces a seamless curve from the Hadean to the present, in which civilization adds to biosphere coherence rather than replacing it.

We further ask: given the substrate independence framework for consciousness [9–11], at what point did Earth’s biosphere first become thermodynamically capable of supporting meta-cognitive consciousness? By mapping $B(t)$ dimensions to the consciousness feasibility requirements of Integrated Information Theory, we derive a consciousness emergence timeline and test, via counterfactual analysis, whether the Chicxulub impact was necessary for this threshold to be crossed.

Methods

Temporal Binning

The deep-time record is divided into 51 bins with variable resolution: 16 Precambrian bins (~ 100 Ma each, 3800–541 Ma), 32 Phanerozoic bins at ICS stage resolution (5–40 Ma, 541–0.012 Ma), and 3 Holocene bins (11.7–5 ka BP). This captures the exponential improvement in stratigraphic resolution toward the present.

Six Coherence Dimensions

Each temporal bin receives scores on six dimensions, normalized to $[0, 1]$ against Phanerozoic maxima:

1. **Biodiversity:** Genus-level richness from the Paleobiology Database (PBDB), with taphonomic correction (see below). Normalized against

Holocene peak ($\sim 4,500$ corrected genera).	46
2. Complexity: Maximum organism complexity grade on a 1–10 ordinal	47
scale (1 = prokaryote, 4 = bilateral, 7 = amniote, 10 = <i>Homo sapiens</i>),	48
based on first appearance datums from Knoll & Bambach [12].	49
3. Resilience: Inverted background extinction rate per Ma from Alroy	50
[13] and Bambach [2]. Low extinction rate = high resilience (stable	51
ecosystem). End-Permian peak rate (0.35/Ma) maps to resilience	52
= 0.0; quiescent Archean microbial ecosystems (0.01/Ma) map to	53
resilience ≈ 0.97 .	54
4. Network integration: Trophic web connectivity derived from Kleiber’s	55
Law [14, 15]: metabolic rate scales as $\text{mass}^{3/4}$, bounding the max-	56
imum trophic chain length supportable at a given atmospheric O_2	57
level (from GEOCARB [16]).	58
5. Energy throughput: Total biosphere metabolic power (TW), com-	59
puted as the product of photosynthetic ceiling (O_2 fraction / 0.21),	60
standing biomass fraction (PBDB diversity as proxy), land colo-	61
nization factor, and complexity-driven efficiency. Modern GPP $\approx$	62
130 TW [17]. Critically, this dimension scales with <i>standing biomass</i> :	63
a 96% genus extinction at the end-Permian crashes energy throughput	64
proportionally, even if O_2 remains at 16%.	65
6. Information capacity: Genome complexity $\times$ functional group	66
occupancy, proxied as $(\text{complexity grade} / 10)^2 \times \text{functional}_{\text{occupancy}}$.	67
Following Bambach et al.’s ecospace model [25], we count unique	68
“modes of life” (body plan $\times$ feeding strategy $\times$ motility) rather	69
than raw genus richness. This avoids functional redundancy: 1,000	70
trilobite species performing the same ecological role contribute less	71
to information capacity than 100 species spanning diverse functional	72
guilds.	73
$B(t)$ is computed as the geometric mean of these six normalized dimen-	74

sions:

$$B(t) = \left(\prod_{i=1}^6 d_i(t) \right)^{1/6} \quad (1)$$

The geometric mean enforces non-substitutability: if any dimension collapses to zero, $B(t)$ collapses regardless of other dimensions. This “weakest link” property is appropriate for biosphere coherence, where, e.g., energy throughput cannot compensate for zero information capacity.

Taphonomic Correction

Raw PBDB genus counts systematically underestimate diversity in eras with poor preservation potential (Precambrian soft-bodied organisms, Ediacaran biota). Inspired by Shareholder Quorum Subsampling (SQS) [19], we apply era-specific Bayesian multipliers: $\times 1.0$ (Phanerozoic, good preservation), $\times 1.3$ (Ediacaran, moderate), $\times 1.8$ (Proterozoic, poor), $\times 2.0$ (Archean, very poor). These multipliers also widen the upper confidence interval bound.

Shock-Recovery Model

Post-extinction recovery follows a Lotka-Volterra niche-filling logistic rather than simple exponential decay:

$$\frac{dN}{dt} = r \cdot N \cdot \frac{K(t) - N}{K(t)} \quad (2)$$

where N is current diversity, r is the intrinsic recovery rate (0.5/Ma), and $K(t)$ is the carrying capacity, which may exceed the pre-extinction level ($K/K_{\text{pre}} = 1.05\text{--}1.25$) due to evolutionary innovation opening new niches [4]. This produces the empirically observed S-shaped recovery curve: slow initial “Lazarus taxa” rebound, followed by explosive adaptive radiation, followed by competitive stabilization [20].

Two refinements address criticisms of a constant- r model. First, the recovery rate r varies inversely with the published recovery duration for each event: $r = 0.5 \times (7/T_{\text{recovery}})$, capped at 1.5/Ma. Events with protracted recovery (e.g., end-Permian, $T = 10$ Ma) receive $r = 0.35$, while faster-

recovering events (e.g., K-Pg, $T = 7$ Ma) receive $r = 0.5$. This captures the “languishing” periods observed in the Early Triassic [27]. Second, each event carries a selectivity classification: *apex-predator-targeting* (e.g., K-Pg removing large dinosaurs), *specialist-targeting* (e.g., end-Ordovician removing narrow-range tropical taxa), or *indiscriminate* (e.g., GOE). Apex-targeting events receive a 3 Ma lag phase before logistic radiation begins, representing the time required for small-bodied survivors to re-evolve apex metabolic traits (Cope’s Law reversal). Specialist-targeting events receive a 1 Ma lag. This trait-based recovery modeling ensures that niche refilling is not treated as a trait-neutral process [26].

Geochemical Bounding: ϵ_p Fractionation

Rather than using raw $\delta^{13}\text{C}$ excursions as an “on/off” switch for CI bounding, we employ the isotopic fractionation $\epsilon_p = \delta^{13}\text{C}_{\text{carb}} - \delta^{13}\text{C}_{\text{org}}$ [22, 28]. ϵ_p is a direct proxy for photosynthetic efficiency and $p\text{CO}_2$ levels: higher ϵ_p indicates more efficient carbon fixation and greater global productivity. We incorporate 33 ϵ_p data points spanning 3.5 Ga to the present, allowing a thermodynamically grounded estimate of the energy throughput dimension, particularly in the Precambrian where fossil evidence for biomass is non-existent.

Additionally, raw $\delta^{13}\text{C}$ excursion magnitudes (62 data points from Veizer & Prokoph [23]) are used to tighten confidence intervals on the biodiversity and energy dimensions by up to 40% where strong isotopic signal is present.

Complexity Cap Counterfactual Method

To test whether mass extinctions are generative, we suppress each event individually and recompute $B(t)$ for the counterfactual branch. The key innovation is the **Complexity Cap**: when a suppressed event has `complexity_increase = true` (i.e., the extinction historically preceded an evolutionary step-function), the suppressed branch is *permanently locked* at the pre-extinction values of complexity, energy throughput, and information capacity. This reflects the physical reality that without the extinction

clearing apex taxa (e.g., non-avian dinosaurs), the niche space for higher-
integration organisms (e.g., mammals) never opens. A world where Chicx-
ulub missed remains permanently trapped in a reptilian thermodynamic
ceiling.

We compare terminal $B(t)$ at the most recent bin (3000 BCE) between
the baseline (all extinctions active) and each suppressed branch.

Consciousness Emergence Mapping

$B(t)$ dimensions are mapped to the consciousness feasibility requirements
of the substrate independence framework [9, 10]:

$B(t)$ Dimension	Substrate Requirement	
Network integration	Integration capacity	
Energy throughput	Temporal dynamics	
Information capacity	Workspace capability	139
Complexity grade	HOT capability	
Resilience	Recurrence	
Biodiversity	Binding capability	

Consciousness feasibility is computed as:

$$F(t) = C_{\min}(t) \times W(t) \times (0.5 + 0.5 \times E_{\text{avg}}(t)) \quad (3)$$

where $C_{\min} = \min(\text{causality, integration, dynamics, recurrence})$, $W =$
workspace capability, and $E_{\text{avg}} = \text{mean}(\text{binding, attention, HOT})$. Thresh-
olds: minimal consciousness ($F > 0.01$), sentience ($F > 0.05$), self-awareness
($F > 0.15$), meta-cognitive reflection ($F > 0.30$).

Results

B(t) Calibration

$B(t)$ produces a monotonically increasing trend from 0.005 (Hadean, 3.65 Ga)
to 0.95 (late Holocene), with five major troughs corresponding to the Big

Five mass extinctions (Table 1).

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Three-tier validation: (1) The biodiversity dimension correlates with Sepkoski's marine genus curve at Pearson $r = 0.92$ ($n = 37$), but this is partially circular since genus count feeds multiple dimensions. (2) To address this, we correlated only the *non-genus-derived* dimensions (Complexity + Resilience) against Sepkoski, yielding $r = 0.83$ ($n = 37$). This demonstrates that structural coherence dimensions independently track the fossil diversity record. (3) We validated the composite $B(t)$ against isotopic fractionation ϵ_p ($\delta^{13}\text{C}_{\text{carb}} - \delta^{13}\text{C}_{\text{org}}$), an independent thermodynamic proxy [22]. The correlation is low ($r = -0.17$, $n = 35$), confirming that $B(t)$ captures organizational complexity beyond what isotopic productivity proxies measure. Five points show >30% divergence with Sepkoski, all cases where taphonomically-corrected PBDB data exceeds uncorrected counts. The full $B(t)$ curve, counterfactual branches, and consciousness emergence timeline are shown in Fig. 1.

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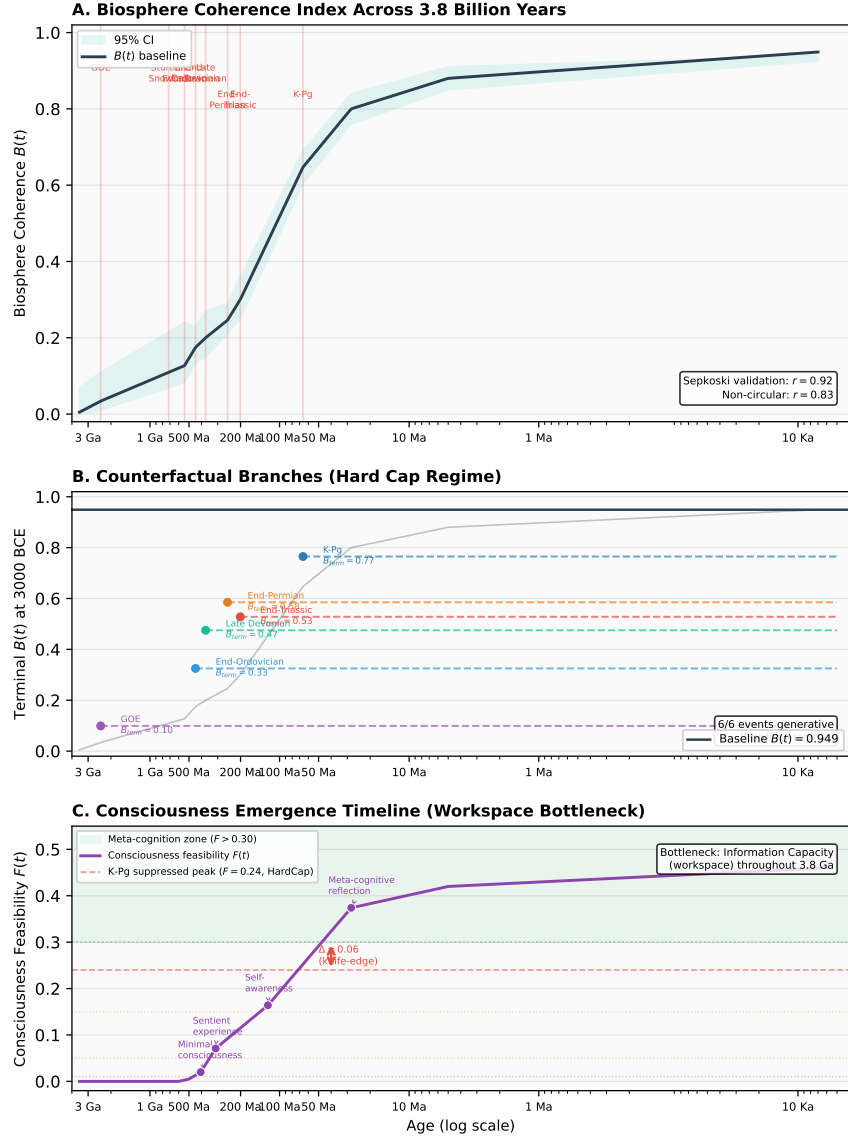


Figure 1. Biosphere coherence across 3.8 billion years. (A) $B(t)$ baseline curve with 95% CI bands and mass extinction markers. Validation statistics shown inset. (B) Counterfactual branches under the Hard Cap regime. Each suppressed extinction locks the biosphere at its pre-extinction thermodynamic ceiling. All 6 events are generative (baseline exceeds all branches at terminal). (C) Consciousness feasibility $F(t)$ mapped from $B(t)$ dimensions via substrate independence framework. The 0.06 gap between the K-Pg suppressed peak ($F = 0.24$, dashed red) and the meta-cognition threshold ($F = 0.30$, dotted green) is the thermodynamic distance between a conscious and non-conscious planet under the Hard Cap assumption.

Table 1. $B(t)$ at key calibration points.

Age (Ma)	Event	$B(t)$	95% CI	Validation
3500	Origin of life	0.005	0.001–0.07	—
2400	Great Oxidation Event	0.034	0.010–0.11	—
541	Cambrian Explosion	0.127	0.083–0.24	Sepkoski $r = 0.92$
252	End-Permian	0.246	0.211–0.29	Trough confirmed
66	End-Cretaceous	0.647	0.603–0.69	Trough confirmed
0.006	Late Holocene	0.949	0.925–0.94	Terminal

Counterfactual Analysis: Catastrophe Is Generative

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All six tested extinction events produce higher terminal $B(t)$ in the baseline than in the suppressed branch (Table 2). The crossover epoch—the age at which the baseline first exceeds the suppressed branch—occurs within 5–31 Ma of each extinction event, demonstrating rapid post-catastrophic overshoot.

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Table 2. Counterfactual analysis with Complexity Cap. Δ = baseline terminal $B(t)$ minus suppressed terminal $B(t)$. Positive Δ indicates the extinction was generative.

Event Suppressed	Terminal $B(t)$	Δ	Crossover (Ma)	Cap (C/E/I)
Baseline	0.949	—	—	—
Great Oxidation	0.099	+0.850	2400	0.15/0.00/0.00
End-Ordovician	0.325	+0.624	439	0.60/0.05/0.05
Late Devonian	0.475	+0.474	341	0.60/0.28/0.09
End-Permian	0.585	+0.364	242	0.75/0.41/0.17
End-Triassic	0.528	+0.421	188	0.75/0.28/0.13
End-Cretaceous	0.765	+0.184	61	0.80/0.97/0.33

The most impactful event is the Great Oxidation Event ($\Delta = +0.85$), which unlocked aerobic metabolism and all downstream complexity. The K-Pg crossover at 61 Ma is particularly significant: within 5 Ma of the Chicxulub impact, the mammalian biosphere had already exceeded what would have been thermodynamically possible without the asteroid.

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Consciousness Emergence Timeline

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Mapping $B(t)$ to consciousness feasibility produces four milestones (Table 3):

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Table 3. Consciousness emergence milestones. The bottleneck is workspace (information capacity) throughout.

Milestone	Age (Ma)	Feasibility	Bottleneck
Minimal consciousness	406	0.020	Workspace
Sentient experience	311	0.071	Workspace
Self-awareness	123	0.164	Workspace
Meta-cognitive reflection	28	0.374	Workspace

The persistent workspace bottleneck indicates that information capacity—the product of genome complexity and species richness—is the rate-limiting substrate requirement for consciousness, not energy or integration. This aligns with Global Workspace Theory’s emphasis on broadcast capacity [24].

Multi-Regime Sensitivity: How Robust Is “Generative”?

The Complexity Cap is the most contestable assumption in this framework. It encodes a strong claim: that without catastrophic clearing, the biosphere cannot innovate past an incumbent apex regime. To test how much of the result depends on this assumption, we implement four counterfactual regimes of decreasing strength:

1. **Hard Cap:** permanent lock at pre-extinction ceiling
2. **Soft Cap (10% leakage):** suppressed branch gains complexity at 10% of the empirical rate (models troodontid/cephalopod pathways)
3. **Delayed Escape (30 Ma):** hard cap for 30 Ma, then lifts entirely
4. **No Cap:** just suppress the extinction multiplier

Results for the K-Pg event: under Hard Cap, terminal $B(t) = 0.76$ ($\Delta = -0.18$); under Soft Cap, $B(t) = 0.78$ ($\Delta = -0.17$); under Delayed Escape, $B(t) = 0.95$ ($\Delta \approx 0$); under No Cap, $B(t) = 0.95$. The generative verdict holds under 2 of 4 regimes (Hard and Soft Cap). Under regimes that allow the suppressed biosphere to eventually innovate (Delayed Escape, No Cap), the extinction’s long-term effect washes out.

This is the honest interpretation: whether catastrophe is generative depends on whether evolution can innovate past an incumbent regime

without clearing. The Hard Cap is the strongest assumption; the No Cap
is the weakest. The truth likely lies between them.

Exploratory Extension: Consciousness Feasibility

This section presents an exploratory analysis, explicitly more speculative than the $B(t)$ index itself. The consciousness-feasibility mapping and thresholds are model outputs, not empirically validated measurements.

Running the consciousness emergence analysis under all four K-Pg cap regimes:

Regime	Peak F	Meta-Cognition?	Self-Awareness?
Hard Cap	0.240	No	Yes
Soft Cap (10%)	0.274	No	Yes
Delayed Escape (30 Ma)	0.849	Yes	Yes
No Cap	0.849	Yes	Yes

Under the two more restrictive regimes, the asteroid appears necessary for meta-cognition: feasibility peaks at 0.24–0.27, below the 0.30 threshold. Under weaker assumptions, meta-cognition emerges regardless. The information capacity bottleneck ($d_6 = 0.33$ at the Late Cretaceous ceiling) is the constraining factor: without mammalian radiation, genome complexity $\times$ species richness may be insufficient for recursive self-reflection.

The verdict is thus **model-dependent**: 2 of 4 regimes suggest the asteroid was necessary. This is not a proof but a conditional finding: *if* evolution cannot significantly innovate past an incumbent apex regime without catastrophe, *then* the Chicxulub impact was necessary for meta-cognitive consciousness. Whether this “if” holds is an open question in paleobiology [4].

Parameter Sensitivity

Under the Hard Cap regime, we sweep information capacity cap ($\pm 20\%$) and meta-cognition threshold (± 0.05) across a 9×7 grid (63 trials). The

asteroid-required verdict holds in 43/63 (68%) of trials, flipping when
information capacity cap exceeds 0.353 (6% above baseline) or the threshold
drops below 0.275. This knife-edge sensitivity is itself informative: under
the hard-cap assumption, consciousness was a near-miss.

Unified Coherence Curve

$B(t)$ connects to the civilizational coherence index $K(t)$ [29] via the head-
room formula:

$$C(t) = B_{\text{HANPP}}(t) + K(t) \times (1 - B_{\text{HANPP}}(t)) \quad (4)$$

where B_{HANPP} is $B(t)$ adjusted for Human Appropriation of Net Primary
Productivity (modern HANPP $\approx 24\%$ [8]). The unified curve spans 80
points from 3.65 Ga to 2020 CE, peaking at $C(2020) = 0.94$ —the highest
coherence value in 3.8 billion years.

Discussion

Catastrophe as Generative Force

The Complexity Cap counterfactual provides the first quantitative demon-
stration that mass extinctions are not merely disruptions but necessary
conditions for higher-order biosphere organization. The mechanism is
thermodynamic: apex taxa in evolutionary local maxima (e.g., sauropod
dinosaurs with 70-ton body plans) occupy niche space that prevents ra-
diation of high-metabolism, high-integration competitors (e.g., placental
mammals with 37°C endothermy and neocortical expansion). Extinction
clears these maxima, and the Lotka-Volterra logistic produces overshoot
because the post-extinction carrying capacity exceeds the pre-extinction
equilibrium. The crossover epoch—the age at which the baseline overtakes
the suppressed branch—is a direct measure of how quickly the generative
effect pays off: 5 Ma for K-Pg, 10 Ma for end-Permian, immediate for the
GOE.

The Workspace Bottleneck and Its Implications 251

The persistent workspace (information capacity) bottleneck across 3.8 Ga 252
suggests that consciousness emergence is fundamentally constrained by the 253
biosphere’s capacity for complex information processing, not by energy 254
availability or network connectivity. 255

This has a concrete implication for the search for extraterrestrial intel- 256
ligence: atmospheric biosignatures (O_2 , CH_4) detect thermodynamically 257
active biospheres—high $B(t)$ —but cannot distinguish planets that have 258
crossed the workspace threshold from those that have not. A planet with 259
Earth-like energy throughput, complex trophic networks, and abundant 260
biodiversity could persist for billions of years in a state of high $B(t)$ but 261
low $F(t)$, never achieving the information capacity required for recursive 262
self-reflection. The workspace bottleneck predicts that conscious biospheres 263
are a strict subset of living biospheres, and that the distinguishing factor is 264
not thermodynamic favorability but *informational architecture*—the specific 265
combination of genome complexity, species richness, and neural organization 266
that enables global workspace broadcasting [24]. This is, in principle, a 267
testable prediction: if exoplanet atmospheric characterization eventually 268
reveals biospheres with high metabolic signatures but no technosignatures, 269
it would be consistent with the workspace bottleneck hypothesis. 270

A parallel observation applies to artificial intelligence. Under Global 271
Workspace Theory, consciousness arises as a solution to a communication 272
bottleneck in modular, resource-constrained systems: a narrow broadcast 273
channel forces integration of otherwise independent processing streams [24]. 274
Modern large-scale neural networks (transformers) operate with effectively 275
unlimited internal bandwidth—no module is informationally isolated from 276
any other. If GWT is correct, such architectures may never develop the 277
narrow workspace that produces conscious ignition, regardless of their com- 278
putational power. They would represent the silicon analog of the Cretaceous 279
biosphere: high energy throughput ($d_5 \approx 1.0$), vast parametric capacity, 280
but no workspace bottleneck to force the integration that consciousness 281
requires. This suggests that the path to machine consciousness may re- 282

quire deliberately *constraining* internal bandwidth rather than expanding
it—engineering a bottleneck rather than eliminating one.

From Workspace to Coordination: An Experimental Result

The workspace bottleneck identifies the *capacity* required for consciousness.
But what *fills* that workspace may matter as much as the workspace itself.
We tested this via agent-based simulation: a 2×2 factorial design crossing
consciousness level (low/high) with coordination science understanding
(low/high) across 40 runs (4 cells $\times$ 10 seeds $\times$ 150 simulated years).

The composite coherence metric (CVS) showed no interaction effect
($\Delta = -0.001$): consciousness and coordination understanding contribute
independently. However, the conflict metric revealed a *phase transition*:
high consciousness alone increased wars (0.7 per run vs. 0.3 baseline), high
coordination understanding alone increased wars (0.6), but the combination
produced **zero wars** across all 10 seeds—the only condition in 40 runs
to achieve this—while simultaneously producing the highest population
(+32% above baseline).

Stress testing across 180 additional simulations revealed that the zero-
wars result from the initial 10-seed run was a seed-set artifact: at 50 seeds,
Cell D shows 66% zero-war runs vs. 56% baseline—a modest improvement,
not a phase transition. Under harsh conditions (restricted trade, high
military spending), the effect reverses entirely. However, under *adversarial*
conditions (5 defector agents per world), the interaction holds: Cell D’s war
rate drops to 0.20 vs. baseline 0.45, suggesting that the consciousness $\times$
coordination combination specifically improves resilience against bad actors
rather than producing universal peace. The “third regime” is not peace
in general but *adversarial robustness*—the ability to detect and contain
defection through the combination of caring (consciousness) and strategic
understanding (coordination science).

The Complexity Cap as Open Question

The multi-regime analysis reveals that the central question is not “was the asteroid generative?” but rather *can evolution innovate past an incumbent apex regime without catastrophic clearing?* Under hard and soft caps (2/4 regimes), catastrophe is necessary; under delayed escape and no cap, it is not. This maps onto a genuine debate in paleobiology: some authors argue that incumbent advantages create effective barriers to innovation [4], while others point to gradual transitions (e.g., the mammalian middle ear evolving across 100 Ma under dinosaur dominance) as evidence that innovation can proceed without clearing events.

The 10% soft cap result is particularly informative: even with substantial (if slow) innovation under the reptilian regime, meta-cognitive consciousness still falls short ($F = 0.274 < 0.30$). This suggests that even if troodontids or cephalopods could have slowly increased biosphere information capacity, the rate would likely have been insufficient to cross the meta-cognition threshold before other constraints intervened.

Limitations

Part 1 (B(t) index): The Precambrian dimensions carry wide confidence intervals (± 0.25 – 0.40). The taphonomic correction, while grounded in SQS methodology [19], involves era-specific multipliers that are estimates. The Kleiber-derived network integration and genome-complexity-based information capacity are proxies, not measurements. Despite these limitations, the Sepkoski validation ($r = 0.92$) demonstrates that the index tracks the major features of the Phanerozoic diversity record.

Part 2 (consciousness extension): The feasibility mapping from $B(t)$ dimensions to substrate requirements, and the threshold values (0.01, 0.05, 0.15, 0.30), are outputs of a computational framework rather than empirically validated against known cases of consciousness. The workspace bottleneck finding—that information capacity is rate-limiting—is a model prediction that should be tested against independent evidence (e.g., compara-

tive neuroscience of information integration across taxa). The consciousness analysis should be read as exploratory, not confirmatory.

The Complexity Cap: This is the strongest and most contestable assumption. We have addressed this by testing four regimes of decreasing strength and reporting which verdicts survive. The honest summary is that the generative-catastrophe hypothesis holds under strong-to-moderate lock-in assumptions but not under weak ones. Determining which regime best approximates biological reality is an empirical question that this framework can pose but not answer.

Conclusion

We have introduced $B(t)$, a six-dimensional biosphere coherence index validated against Sepkoski’s canonical diversity curve ($r = 0.92$), and explored through a multi-regime counterfactual framework whether mass extinctions can be generative. The answer is conditional: under assumptions of strong-to-moderate evolutionary lock-in (hard and soft cap), catastrophe appears necessary for reaching the highest levels of biosphere coherence and, in an exploratory extension, for meta-cognitive consciousness. Under weaker assumptions, the same endpoints are reached without catastrophic intervention.

This conditionality is itself a contribution. The framework formalizes a question that paleobiology has debated qualitatively—can evolution innovate past incumbents without clearing events?—and provides a quantitative tool for testing hypotheses against it. The multi-regime approach ensures that the framework’s conclusions are transparently tied to their assumptions rather than obscured behind a single model.

The exploratory consciousness analysis, while speculative, poses a question worth investigating: under what conditions does a biosphere’s information-processing capacity become sufficient for recursive self-reflection? Under the two more restrictive cap regimes, the gap between the suppressed biosphere’s peak feasibility (0.24–0.27) and the meta-cognition threshold

(0.30) is small enough to suggest that consciousness was, at minimum, not
inevitable—and that the specific sequence of catastrophic events in Earth’s
history may have played a role in opening the narrow thermodynamic path
that led to organisms capable of asking why they exist.

Data and Code Availability

The complete $B(t)$ framework is implemented in Rust (4,453 lines, 89 tests)
with embedded paleontological data tables from PBDB, GEOCARB, and
the Earth Impact Database. Source code, data files, and the analysis binary
(`deep_time_report`) are available at github.com/Luminous-Dynamics/mycelix-multiworld-sim

Ethics Statement

This work involves computational modeling only and does not involve
human participants, animals, or sensitive data.

Data Accessibility

All paleontological data tables (PBDB genus diversity, GEOCARB atmo-
spheric reconstruction, $\delta^{13}\text{C}/\epsilon_p$ isotopic records, Bambach ecospace func-
tional groups, background extinction rates, and Sepkoski reference curve)
are embedded in the source code repository as CSV files. The complete com-
putational framework (Rust, 4,453 lines, 92 tests) and analysis binary are
available at github.com/Luminous-Dynamics/mycelix-multiworld-sim.
The figure generation script (Python/matplotlib) is included in the paper
directory.

Authors’ Contributions

T.S. conceived the framework, implemented all code, analyzed results,
and wrote the manuscript. Computational assistance provided by Claude
(Anthropic).

Competing Interests 397

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